Use Cases

Real-World Use Cases

Browse practical implementations and scenarios that show how Archimedes enables faster insights and action.

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Adaptive Test with UltraEdge

Watch a comprehensive video demonstration of how Adaptive Test can be implemented with UltraEdge technology to create intelligent, self-optimizing test programs.

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REbin Application based on GDBN Algorithm

Watch a comprehensive video demonstration of how Rebin Application can be implemented with the GDBN technology.

Video: Adaptive Test with UltraEdge - RA_0451

⚡ Adaptive Test and UltraEdge Demo Video

Reference: RA_FAARCH_451

Watch this comprehensive demonstration of how Adaptive Test can be implemented with UltraEdge technology to create smarter, more flexible, and responsive test programs.

Overview

This video demonstrates the complete implementation of Adaptive Test using UltraEdge technology. It showcases how test programs can dynamically adapt to real-time conditions, improving test efficiency and quality without modifying the original test program.

Watch the Demo

0:00

What You'll Learn

In this video demonstration, you'll discover:

1. **Test Time Reduction** - How tests are automatically disabled when they consistently achieve specific yield thresholds (e.g., 98%)
2. **Online Analysis** - Automatic triggering of diagnostic flows when tests fail too frequently
3. **Quality Improvement** - Dynamic adaptation of test limits based on ongoing test results
4. **Secure Algorithm Deployment** - How to protect intellectual property using encrypted Docker containers on UltraEdge

Key Features Demonstrated

Intelligent Test Optimization

The system analyzes current results in real-time to make intelligent decisions:

- Automatically disables tests that consistently pass to save time
- Triggers diagnostic flows when failure rates exceed thresholds
- Adjusts test limits dynamically to match device characteristics

Secure Infrastructure

The video showcases:

- **Encrypted Docker Deployment** - Secure packaging and deployment of proprietary algorithms

- **IP Protection** - Safeguarding sensitive logic even in third-party test environments
- **Zero Test Program Modification** - Seamless integration through AMP without code changes

Architecture Components

The demonstration covers the complete system architecture:

- **UltraEdge Computer** - Hosts and executes the encrypted algorithm
- **AMP Integration** - Enables secure communication with the test program
- **Real-time Data Exchange** - Captures test data and applies adaptive decisions
- **Demo Application** - Visualizes the entire process across four dedicated sections

Video Content Structure

1. **Introduction** - Overview of Adaptive Test capabilities
2. **Algorithm Explanation** - Detailed walkthrough of the decision logic
3. **Infrastructure Setup** - UltraEdge deployment and security measures
4. **Live Demonstration** - Step-by-step execution showing:
    - Test program loading
    - UltraEdge deployment
    - Loop execution and real-time adaptation
    - Test activation and limit updates
    - STDF file validation
5. **Technical Implementation** - Code examples and integration points

Technical Environment

The video demonstration uses the following technical environment:

Component	Version
Language	C# & Python
Framework	.NET 8
Environment	Windows 10
Excel	2016 (64-bit)
IG-XL	10.60.03

Note: These specifications reflect the environment used in the recorded demonstration. The implementation can be adapted to work with other compatible versions.

Related Resources

For hands-on experience and detailed instructions:

- **RA_468: Implement DAS in C#** - Technical guide for DAS implementation
- **RA_465: Adaptive Commands in Python** - How to send adaptive commands

Implementation Guide

To implement a similar system in your environment:

1. Review the reference architectures listed above
2. Set up your UltraEdge infrastructure
3. Configure AMP for test program communication
4. Develop and encrypt your adaptive algorithm
5. Deploy and test the complete solution

Next Steps

- Watch the video to understand the complete workflow
- Contact the team to discuss implementation for your use case

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💡 Questions or Need Support?

Looking to integrate this approach into your test strategy? Contact our team to explore how Adaptive Test with UltraEdge can support your roadmap.

Video: Adaptive Test with UltraEdge - RA_0476

⚡ GDBN application with Rebin Archimede Application

Reference: RA_FAARCH_476

Watch this comprehensive demonstration of how we can implement GDBN thanks to the Rebin Archimedes Application

Overview

This video demonstrates the complete implementation of Rebin application using GDBN Algorithm. It showcases how Rebin Application can dynamically adapt to the results previously computed from the test program, improving test quality without modifying the original test program.

Watch the Demo

0:00

What we will demonstrate

In this video demonstration, you'll discover the GDBN purpose, which is used in order to demonstrate the power of the Rebin system of Archimedes:

Let me introduce you to GDBN — which stands for Good Die in a Bad Neighborhood. It’s a quality assurance methodology developed for semiconductor manufacturing. The key idea is this: sometimes, a die tests as 'good' — it passes all functional checks. But if that die is physically located in a region surrounded by many failing dies, we have to question its reliability. It may be a false positive. So GDBN doesn’t just look at a die in isolation — it considers the neighborhood. If you're surrounded by trouble, chances are you're in trouble too. This method allows us to flag and reject potentially risky dies before the test ends. The goal is to improve the overall reliability of shipped devices by reducing the chance of marginal parts escaping.”

Key Features Demonstrated

Intelligent Algorithm

This video illustrates a concrete implementation of GDBN in our test flow. We have a typical test sequence — starting from the beginning of the test, running through multiple test steps, and then finishing with final binning. But GDBN introduces something smart between the last test and final binning — a heuristic evaluation step we call 'Pre Test End'. Look at the top-right wafer map: Devices 6 and 10 have initially passed all tests and were marked as good — you see them in green. However, they’re completely surrounded by failed dies — that’s a red flag. This is what we call a bad neighborhood. So before we assign the final binning, GDBN steps in. It questions: “Are they really good devices, or false positives?” Using the GDBN heuristic, we proactively reassign their binning and mark them as failed — as shown in the bottom wafer map. ➡️ This process ensures we catch potential quality risks before final binning, even if individual tests passed. This demonstrates how spatial context enhances quality screening effectiveness through intelligent analysis."

Secure Infrastructure

The video showcases:

- **Implementation of simple Rebin algorithm** - We will demonstrate how to change the binning from an external process
- **How Rebin application will work** - We will demonstrates how the different pieces are working together
- **Monitoring** - We will show the different monitoring put in place while rebin is activated

Architecture Components

The demonstration covers the complete system architecture:

- **Rebin system** - A Rebin application able to change the binning on the fly from external process
- **Demo Application** - Visualizes the entire process across different dedicated sections

Video Content Structure

1. **Introduction** - Overview of Rebin Test capabilities
2. **What is GDBN** - Overview of the GDBN technology
3. **Example of GDBN** - A view of the GDBN system
4. **Why does GDBN Matter** - GDBN can improve a test reliability
5. **Scenario Implemented** - Description of the scenario implemented
6. **Demos**
    - Test program loading
    - Rebin connection
    - Test Starting
    - GDBN implementation
    - DAS Messages for Binning
    - STDF file validation

Related Resources

For hands-on experience and detailed instructions:

- **RA_468: Implement DAS in C#** - Technical guide for DAS implementation
- **RA_465: Adaptive Commands in Python** - How to send adaptive commands

Next Steps

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